SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with "A+" Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation:R26	**I/IV-B.Tech. I -Semester**

ELECTRICAL & ELECTRONICS ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2610001	English for Technical Communication	**HS**	3	0	0	3	40	60	100
A2610002	Linear Algebra and Calculus	**BS**	3	0	0	3	40	60	100
A2610201	Physics for Electrical Systems	**BS**	3	0	0	3	40	60	100
A2610202	Basic Electrical Circuits	**ES**	3	0	0	3	40	60	100
A2610008	Design Thinking and Innovation	**ES**	0	1	3	2.5	50	50	100
A2610009	AI Tools and Applications	**ES**	0	1	3	2.5	50	50	100
A2610203	Physics for Electrical Systems lab	**BS**	0	0	3	1.5	50	50	100
A2610015	IT Workshop	**ES**	0	0	3	1.5	50	50	100
A2610204	Basic Electrical Circuits Lab	**ES**	0	0	3	1.5	50	50	100
A2610018	NSS, Sports & Yoga	**MC**	0	0	2	0	50	-	50
		TOTAL	**12**	**2**	**17**	**21.5**	**460**	**490**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610001	HS	3	--	--	3	40	60	3 hrs

ENGLISH FOR TECHNICAL COMMUNICATION

(Common to all Branches)

Course Objectives:

1.	Enable students identify the context and thematic information from social, academic, and workplace dialogues and texts.
2.	Develop students' ability to use appropriate grammatical structures and vocabulary.
3.	Equip students with contextual vocabulary for writing.
4.	Familiarize students with reading strategies to comprehend and interpret texts.
5.	Develop students' ability to demonstrate appropriate social media etiquette.

Course Outcomes

Sl. No	Outcome	Knowledge Level
1.	Identify the thematic information from transactional dialogues and texts to construct coherent written and spoken discourses.	K2
2.	Apply appropriate grammatical structures and vocabulary to summarize texts on the related themes.	K3
3.	Apply contextual vocabulary to paraphrase texts and write letters.	K3
4.	Develop reading strategies to comprehend and interpret various texts for a global understanding.	K3
5.	Construct book/film reviews and demonstrate appropriate social media etiquette in writing, employing sentence synthesis techniques.	K3

SYLLABUS

UNIT-I (10Hrs)	**Effective Communication: Exploring Channels and Methods** **Vocabulary:** Roots, Prefixes, and Suffixes; Homonyms **Grammar:** Articles, Prepositions, Tenses. **Listening:** Listening for main ideas and supporting details, Note-taking **Speaking:** Introducing self and others, Describing people and objects, JAM **Reading:** Definition and Importance, Skills and Sub-Skills of Reading **Writing:** Cohesion and Coherence, Paragraph Writing
UNIT-II (10 Hrs)	**Eco-Consciousness: Negotiating for a Sustainable Future** **Vocabulary:** Contractions, Phrasal Verbs, Idioms & Phrases **Grammar:** Subject-Verb Agreement, Conditional Sentences **Listening:** Listening to various texts and comprehension **Speaking:** Role-plays, Extempore **Reading:** Reading for Inference, Point of View, Tone and Perspective **Writing:** Note-making, Summarizing, Paraphrasing (at basic level)
UNIT-III (10 Hrs)	**Disruptive Technologies: Ethical Implications and Future Prospects** **Vocabulary:** One-Word Substitutes, Commonly Confused Words **Grammar:** Degrees of Comparison, Active and Passive Voice, Question Tags

	Listening: Listening to stories, news, documentaries, and discussions **Speaking:** Debate, Group Discussions **Reading:** Reading the extracts/articles/ essays of the newspapers, journals, and literary texts. **Writing:** Paraphrasing (at advanced level), Letter Writing
UNIT-IV (10 Hrs)	**Entrepreneurship and Start-Up Culture: Emerging Trends** **Vocabulary:** Business/ Profession/ Career/ Job-related Vocabulary **Grammar:** Sentence Synthesis, Simple, Compound and Complex Sentences **Listening:** Listening to debates and discussions **Speaking:** PowerPoint Presentations by Individuals and Teams **Reading:** Reading general texts for global understanding **Writing:** Email Writing, Data Interpretation, Report Writing
UNIT-V (10 Hrs)	**Ethical Leadership: Integrating Social Responsibility and Core Values** **Vocabulary:** Advanced Vocabulary for GRE/TOEFL/ IELTS/CAT **Grammar:** Sentence Synthesis, Para Jumbles **Listening:** Listening to talks/speeches by eminent personalities **Speaking:** Team Discussions and Presentations **Reading:** Reading specific texts for global comprehension **Writing:** Book/Film Reviews, Social Media Etiquette
Textbook:	
1.	Chekuri, T., & Perumalla, R. K. (2026). NextGen English for engineers: A course book for technical English communication. Cengage.
Reference Books:	
1.	Technical Communication (2nd Ed.) by M. Raman & S. Sharma. Oxford University Press, 2012.
2.	Effective Technical Communication by M.A. Rizvi, Tata McGraw-Hill, 2005.
3.	English for Engineers and Technologists, Orient Blackswan 2010 (Combined edition, Vol.1 &2).
4.	The Oxford Guide to Writing and Speaking by John Seely, Oxford University Press, 2004.
e-Resources	
1.	eGyanKosh. (n.d.). National digital repository. https://egyankosh.ac.in/
2.	Read Theory. (n.d). Adaptive reading comprehension practice. https://readtheory.org/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610002	BS	3	0	--	3	40	60	3 Hrs.

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Course Objectives: Students are expected to learn

1.	Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations.
2.	Eigen values, Eigen vectors and quadratic forms.
3.	Problems and applications of Mean value theorems
4.	Application of partial differentiation for determining maxima/ minima of functions.
5.	Concepts of double, triple integrals and its applications.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Solve** a system of linear equations by using direct and indirect methods.	K3
2.	**Determine** Eigen values and Eigen vectors, execute matrix Diagonalization and reduce Q.F.'s to canonical forms.	K3
3.	**Apply** mean value theorems to analyze function behaviour and solve real life problems.	K3
4.	**Apply** the concepts of partial differentiation in various engineering applications.	K3
5.	**Apply** the concepts of multiple integrals to calculate areas, volumes of geometrical regions and in engineering applications.	K3

SYLLABUS

UNIT-I (10Hrs)	**Linear System of Equations:** Rank of a matrix by echelon form, normal form. Inverse of a matrix by Gauss-Jordan method. Consistency of linear equations: Homogeneous and Non-Homogeneous systems. Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods.
UNIT-II (10 Hrs)	**Eigen Values, Eigen Vectors and Quadratic Forms:** Eigen values and Eigen vectors of a matrix, their properties, Cayley-Hamilton Theorem (without proof), Inverse and powers of a matrix by Cayley – Hamilton theorem, Diagonalization of a matrix, Quadratic forms and Nature of the Quadratic Forms, Reduction of a Quadratic form to canonical form by Orthogonal Transformation and Singular Value Decomposition.
UNIT-III (10 Hrs)	**Single Variable Calculus:** Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem, Geometrical interpretation of these theorems, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems (without proof) with Lagrange's form of remainder, Taylor's and Maclaurin series.

UNIT-IV (10 Hrs)	**Partial differentiation and Applications:** Functions of several variables, Continuity and Differentiability, Partial derivatives, Euler's theorem, Total derivative, chain rule, Taylor's and Maclaurin's series expansion of function of two variables, Jacobians, Functional dependence, Maxima and Minima of function of two variables without constraints and with constraints - Method of Lagrange multipliers.
UNIT-V (10 Hrs)	**Multiple Integrals:** Double integrals, change of order of integration, triple integrals, change of variables from Cartesian to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	ThomasCalculus,GeorgeB.Thomas,MauriceD.WeirandJoelHass,PearsonPublishers,2018, 14thEdition.
2.	Higher Engineering Mathematics, B.V.Ramana, McGraw HillEducation,2017.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th edition.
5.	Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications,2014, Third Edition (Reprint 2021).
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty Two EditionReprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.

e-Resources	
1.	https://en.wikipedia.org/wiki/System_of_linear_equations
2.	https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors
3.	https://www.math.hmc.edu/calculus/tutorials/eigenstuff/
4.	https://en.wikipedia.org/wiki/Quadratic_form
5.	https://en.wikipedia.org/wiki/Calculus
6.	https://en.wikipedia.org/wiki/Partial_derivative
7.	https://www.whitman.edu/mathematics/calculus_online/section14.03.html
8.	https://en.wikipedia.org/wiki/Multiple_integral
9.	http://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610201	BS	3	--	--	3	40	60	3 Hrs.

PHYSICS FOR ELECTRICAL SYSTEMS

(For EEE)

Course Objectives:

1.	To provide knowledge on wave optics concepts and their applications.
2.	To understand the basic concepts of Laser light sources and working principle of optical fibers and identify the engineering applications of lasers and optical fibers in communication.
3.	To familiarize students with concepts of dielectric and magnetic materials and their applications in energy storage devices.
4.	To introduce the electrical properties of metals and semiconductors with relevant engineering applications.
5.	To equip students, the fundamentals of quantum mechanics and quantum computing with secure communication concepts.

Course Outcomes: After successful completion of this course, the student will be able to:

S.No.	Outcome	Knowledge Level
1.	Apply the principles of interference, diffraction, and polarization with basic optical analysis in engineering applications.	K3
2.	Analyze the working of lasers, fiber optics, and advanced sensing systems in optical communication and signal processing techniques.	K4
3.	Analyze the significant concepts of dielectric and magnetic materials that leads to potential applications in the emerging micro devices.	K4
4.	Analyze the electrical properties of metals and comprehend use of semiconductors and their Applications.	K4
5.	Apply of the concepts and principles of Quantum Mechanics relevant to engineering applications such as Quantum Computing for secure communication system.	K3

SYLLABUS

UNIT-I (10Hrs)	**APPLIED WAVE OPTICS:** **Interference:** Superposition Principle – Coherence – Types of Interference – Thin-film interference (reflection geometry) – Newton's rings – Determination of wavelength and refractive index. **Diffraction and Optical Surveying:** Fresnel and Fraunhofer diffraction – Fraunhofer's diffraction due to single slit (quantitative), double slit & N-slits (qualitative) – Diffraction Grating and grating spectrum – Dispersive and Resolving power of a grating (qualitative) . **Polarization:** Types of polarization – Nicol's prism – Half-wave and Quarter-wave plates.

UNIT-II (9 Hrs)	**LASERS AND OPTICAL FIBERS:** **Lasers:** Principles of Absorption – Spontaneous and Stimulated Emission – Population Inversion – Three level and Four level laser systems – Construction and working of Ruby and He-Ne Laser – Applications (qualitative). **Fiber Optics:** Propagation mechanism – Numerical Aperture – Acceptance angle – Types of optical fibres – Classifications and Modes of optical fibres.
UNIT-III (9 Hrs)	**DIELECTRICS & MAGNETIC MATERIALS:** **Dielectric Materials:** Introduction – Basic definitions – Types of polarizations -– Electronic and Ionic polarizations (quantitative)- Internal field in solids – Clausius-Mossotti relation. **Magnetic Materials and Energy Conversion:** Introduction and classification – Dia, Para, Ferro, Ferri and Anti-Ferro – Hysteresis or B-H curve of ferromagnetic materials, Soft & Hard magnetic materials.
UNIT-IV (10 Hrs)	**ELECTRICAL PROPERTIES OF METALS & SEMICONDUCTORS:** **Electrical Properties of Metals:** Classical free electron theory – merits and demerits, Quantum free electron theory – equation for electrical conductivity – Fermi-Dirac statistics – Fermi level and variation with temperature. **Semiconductors:** Formation of energy bands – Classification of solids - Intrinsic semiconductors – Density of charge carriers – Electrical conductivity – Fermi level – extrinsic semiconductors: density of charge carriers (qualitative) – dependence of Fermi energy on carrier concentration and temperature – Drift and diffusion currents – Einstein's equation – Hall effect – Hall coefficient – Applications of Hall effect.
UNIT-V (10 Hrs)	**INTRODUCTION TO QUANTUM MECHANICS & QUANTUM COMPUTING:** **Quantum Mechanics:** de Broglie concept of matter waves – Heisenberg's uncertainty principle – Wave function – Postulates of quantum mechanics – Schrödinger's time dependent and time-independent wave equations -–Particle in a one-dimensional potential well – Quantum tunnelling (basic idea). **Quantum Computing:** Introduction to quantum computing – Classical bits vs qubits – Superposition and entanglement – Basic quantum operations – Hadamard gate – Pauli-X gate – Idea of quantum measurement.
Textbooks:	
1.	Avadhanulu, M.N., Kshirsagar, P.G., & Murthy, T.V.S.A., A textbook of Engineering Physics, S. Chand Publishing (2019)
2.	P.K. PalaniSamy, Engineering Physics, SciTech Publications (2019)
Reference Books:	
1.	D. K. Bhattacharya & PoonamTandon, Engineering Physics, Oxford Press (2015)

2.	S. O. Pillai, Solid State Physics, New Age International(2005).
3.	Nielsen & Chuang, Quantum Computation & Quantum Information, Cambridge (2010).
e-Resources	
1.	NPTEL – nptel.ac.in
2.	MIT OpenCourseWare – ocw.mit.edu
3.	HyperPhysics – hyperphysics.phy-astr.gsu.edu
4.	Feynman Lectures – feynmanlectures.caltech.edu
5.	PhET Simulations – phet.colorado.edu
6.	Library of Congress – loc.gov/rr/scitech/selected-internet/physics.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610202	ES	3	--	--	3	40	60	3 Hrs.

BASIC ELECTRICAL CIRCUITS

(For EEE)

Course Objectives: Student will learn about

1.	The various techniques of circuit analysis using resistive circuits.
2.	The fundamentals of AC circuit analysis.
3.	The various theorems in Electrical Circuits.
4.	The magnetically coupled and resonance phenomenon in Electrical Circuits.
5.	The domestic wiring and Electrical safety

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Analyse DC resistive circuits using circuit laws, network transformations, nodal analysis and mesh analysis techniques.	K4
2.	Analyze single-phase AC circuits using phasor representation, impedance/admittance and steady-state circuit-analysis techniques.	K4
3.	Apply network theorems to obtain responses in DC and AC circuits.	K3
4.	Analyze resonant circuits using bandwidth & quality factor concepts and coupled circuits using dot convention	K4
5.	Apply electrical wiring and protection principles for domestic applications considering load estimation, earthing and protective devices.	K3

SYLLABUS

UNIT-I (10 Hrs)	**DC Circuit Analysis** Circuit elements – Passive elements: R, L & C and their v-i relations - Active Elements: Voltage and Current Sources (Dependent and Independent) – Kirchoff's Laws. Analysis of DC resistive circuits: Single node-pair and single loop circuits, Source combinations, Resistances combinations, Voltage and Current division rules, star/delta transformation, Source transformation, Nodal and Mesh analysis techniques.
UNIT-II (10 Hrs)	**Single Phase AC Circuit Analysis** Characteristics of Sinusoids – Sinusoidal Steady State analysis – Phasor concept – Phasor voltage – current relations for R, L & C elements - Impedance and Admittance concepts - Phasor diagrams - Nodal and Mesh Analysis - Instantaneous power and Average Power - RMS values for periodic voltage and current waveforms - Complex Power.
UNIT-III (10 Hrs)	**Network Theorems** Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem and Reciprocity theorem for DC resistive circuits – Extension to simple AC

	circuits.
UNIT-IV (10 Hrs)	**Resonance and Coupled Circuits** Resonance: Resonance phenomenon - RLC Series and Parallel Resonance - Characteristics of a series and parallel resonant circuits – Bandwidth - Expressions for half power frequencies - Quality Factor - Selectivity. Coupled Circuits: Concept of self and mutual inductance, Dot convention - Coefficient of coupling – Analysis of coupled circuits in Sinusoidal Steady state.
UNIT-V (10 Hrs)	**Electrical Wiring and Safety:** Schematic diagram of Domestic Wiring - Power rating of different household appliances - Load estimation – Digital energy meter - Electricity bill. Electrical Safety: Electric shock, Safety Precautions to avoid shock, Earthing and its types, Protective devices: Fuse, Miniature Circuit Breaker (MCB), and Earth leakage circuit breaker (ELCB) / Residual Current Circuit Breaker (RCCB).
Textbooks:	
1.	Engineering Circuits Analysis, Jack Kemmerly, William Hayt and Steven Durbin, Tata Mc Graw Hill Education, 2020, 9th edition.
2.	Network Analysis, M.E. Van Valkenberg, Pearson Education, 2019, Revised 3rd Edition.
Reference Books:	
1.	Fundamentals of Electrical Circuits, Charles K Alexander and Mathew N.O Sadiku, Mc Graw Hill Education (India), 2022, 7th Edition.
2.	Schaum's Outline of Electric Circuits, by Mahmood Nahvi, Joseph Edminister, McGraw Hill; 7th edition.
3.	S. L. Uppal & G. C. Garg, (2018), Electrical Wiring, Estimating and Costing, (6th edition), Khanna Publishers.
e-Resources	
1.	https://nptel.ac.in/courses/108105159
2.	https://nptel.ac.in/courses/117106108

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610008	**ES**	**--**	**1**	**3**	**2.5**	**50**	**50**	**3 Hrs.**

DESIGN THINKING AND INNOVATION

(Common to AIML, CIC, CIVIL, CSE, CSBS, ECE, EEE, MECH)

Course Objectives: At the end of the course, the teacher should be able to:

1	Introduce students to the principles, mindset and methodologies of **Design Thinking and human-centered problem solving**.
2	Develop students' ability to **observe, empathize, analyze user needs and formulate well-defined engineering and societal problems**.
3	Develop students' **creative and innovative thinking abilities** through systematic ideation, concept generation and evaluation techniques.
4	Enable students to **develop, prototype, test and iteratively improve solutions** using user feedback and appropriate physical/digital tools.
5	Develop students' ability to identify innovation opportunities, create value propositions, understand basic IPR and communicate innovative solutions effectively.

Course Outcomes: After successful completion of the course, student will be able to

S.No	Outcome	Knowledge Level
1	Explain the principles, processes and mindset of Design Thinking for engineering problem-solving and innovation	K2
2	Apply empathy, observation and problem-framing techniques to identify user needs and define engineering problems	K3
3	Apply creative and systematic ideation techniques to generate and select innovative solution concepts	K3
4	Analyze solution concepts and prototypes using user feedback and appropriate design criteria for iterative improvement	K4
5	Analyze the innovation potential of solutions considering value proposition, feasibility, sustainability, societal impact and basic IPR aspects	K4

1	**UNIT I — Design Thinking Foundations & Empathy** Introduction to Design Thinking ; Design thinking mindset and principles ; Design thinking vs. conventional engineering problem-solving ; Human-centered design ; Five-stage design thinking process ; Creativity, curiosity and observation ; Understanding users and stakeholders ; Empathy and user research ; Interviews, observation and contextual inquiry ; Empathy mapping ; Personas and user journeys ; Identifying explicit and latent/unarticulated needs ; Ethical considerations in user research **Practical Activities** • Observation of a real-life problem • Conducting user interviews

	• Preparing an empathy map • Developing a persona • Preparing a user journey map **Unit Outcome** Students will be able to **understand users, identify their needs and translate observations into meaningful design opportunities**
2	**UNIT II — Problem Definition, Analysis & Opportunity Identification** From observations to insights ; Data synthesis and affinity mapping ; Identifying patterns and pain points ; Problem statements ; Point-of-View (POV) statements ; "How Might We" (HMW) questions ; Stakeholder mapping ; 5 Whys and root-cause analysis ; Problem prioritization ; Need–Opportunity mapping ; Problem validation ; Design constraints and specifications ; Social, environmental and sustainability considerations ; Identifying opportunities for innovation **Practical Activities** • Affinity mapping of interview data • Develop three alternative problem statements • Prepare POV and HMW statements • Stakeholder map • Root-cause analysis • Select and validate one problem/opportunity **Unit Outcome** Students will be able to **analyze user research, identify root causes, frame well-defined problems and recognize opportunities for innovative solutions**.
3	**UNIT III — Creativity, Ideation & Concept Development** Creativity and innovation mindset; Divergent and convergent thinking ; Brainstorming and brainwriting ; SCAMPER ; Mind mapping ; Reverse brainstorming ; Analogical thinking ; Biomimicry ; Lateral thinking ; Forced connections ; Morphological analysis ; TRIZ — basic principles ; Individual and collaborative ideation ; AI-assisted ideation: opportunities and limitations ; Idea clustering and screening ; Impact–feasibility matrix ; Concept selection ; Design criteria and decision matrix **Practical Activities** • Generate 20–30 ideas for a selected problem • SCAMPER exercise • Brainwriting exercise • Biomimicry-based ideation • AI-assisted idea generation and critical evaluation • Prepare an idea-selection matrix • Select the top 2–3 concepts **Unit Outcome** Students will be able to **generate diverse solution concepts using creative techniques and systematically select promising ideas based on defined criteria**.

4	**UNIT IV — Prototyping, Experimentation & Validation** Concept development ; Prototype mindset ; Types of prototypes: Paper prototype, Physical prototype, Digital prototype, Storyboard, Role-play, Service prototype ; Low-fidelity and high-fidelity prototypes ; Rapid prototyping ; "Fail fast, learn fast" ; Minimum Viable Prototype (MVP) ; Design experimentation ; Prototype evaluation ; User testing ; Think-aloud method ; Feedback collection ; Usability and functionality testing; Iterative design ; Design critique ; Documentation of iterations ; Use of CAD, simulation and AI-assisted visualization tools **Practical Activities** • Develop a low-fidelity prototype • Prepare a storyboard • Build a physical/digital prototype • Conduct user testing • Collect structured feedback • Modify the prototype • Prepare an iteration log
5	**UNIT V — Innovation, Value Creation & Communication** **Innovation: meaning, characteristics and importance:** Invention vs. innovation ; Incremental, disruptive and radical innovation ; Product, process, service and social innovation ; Technology-driven vs. need-driven innovation ; Sustainable and frugal innovation ; Innovation opportunities and emerging technologies ; From prototype to innovation ; Desirability, feasibility and viability ; Value Proposition ; Value Proposition Canvas ; Introduction to Business Model Canvas ; Customer segments and stakeholder value; Innovation validation; Market/user validation ; Intellectual Property Rights — introduction; Patents, copyright, trademarks and design protection ; Responsible innovation ; Inclusive and sustainable innovation ; Innovation ecosystem and entrepreneurship ; Communicating innovative ideas ; Innovation pitch **Practical Activities** 1. Classify existing products/services as incremental, disruptive or radical innovations. 2. Identify three innovation opportunities from a campus/community problem. 3. Prepare a **Value Proposition Canvas**. 4. Validate the proposed innovation with potential users. 5. Prepare an introductory **Business Model Canvas**. 6. Identify possible IPR protection. Prepare and deliver a **3-minute Innovation Pitch**
Text Books:	
1	Chandan Adhikari, Prabir Kumar Das & Tanay Pramanik, Textbook of Design Thinking and Innovation, S. Chand Publishing, 2026.
2	Shalini Rahul Tiwari & Rohit Rajendra Swarup, Design Thinking: A Comprehensive Textbook, Wiley India, 2024

Reference Books:	
1	Soni, Pavan, Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving, Penguin Random House India.
2	Jauhari, Vinnie & Sudhanshu Bhushan, Innovation Management, Oxford University Press India.
Web References:	
1	Atal Innovation Mission, NITI Aayog – Innovation and Design Thinking Resources. https://aim.gov.in/
2	Institution's Innovation Council (IIC), Ministry of Education, Government of India, https://iic.mic.gov.in/
3	Startup India, Government of India – Innovation and Entrepreneurship Ecosystem, https://www.startupindia.gov.in/
4	National Innovation Foundation – India – Grassroots and Frugal Innovation, https://nif.org.in/
5	Stanford d.school – Design Thinking Bootleg and Design Thinking Resources, https://dschool.stanford.edu/resources/design-thinking-bootleg
6	IDEO.org – Human-Centered Design and Social Innovation Resources, https://www.ideo.org/
7	IDEO U – Design Thinking and Innovation Resources, https://www.ideou.com/pages/design-thinking
8	NPTEL – Online Courses on Design, Innovation, Creativity and Entrepreneurship, https://nptel.ac.in/
9	SWAYAM – Online Courses on Innovation, Entrepreneurship and Design Thinking, https://swayam.gov.in/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610009	ES	--	1	3	2.5	50	50	3Hrs

AI Tools & Applications

(Common to AIDS, CIVIL, CSBS, CSD, CSIT, EEE, IT, MECH)

Course Objectives: To familiarize students with the landscape of AI tools and their underlying working principles.

1	To develop competence in prompt engineering for effective interaction with generative AI systems
2	To enable students to apply AI tools for data analysis, visualization, and interpretation.
3	To train students in using AI-assisted tools for coding, writing, and productivity tasks.
4	To instill awareness of ethical, legal, and societal considerations in AI tool usage.

Course Outcomes:

S. No	Outcome	Knowledge Level
1	Illustrate the categories, capabilities, and limitations of major AI tools (text, image, code, data).	K3
2	Apply the principles of prompt engineering and demonstrate how generative AI models process instructions.	K3
3	Apply AI tools to solve first-year engineering problems in analysis, writing, and coding tasks.	K3
4	Analyze AI-generated outputs for accuracy, bias, and reliability before use.	K4
5	Evaluate the ethical, academic-integrity, and societal implications of AI tool use, justifying appropriate usage choices.	K4

SYLLABUS

1

Introduction to AI and the AI Tools Landscape What is AI, Machine Learning, and Generative AI — basic definitions and how they relate to one another. Core categories of AI tools: conversational/text, code assistants, data-analysis tools, image-generation tools, and writing/productivity tools. How large language models process and generate responses (conceptual, non-mathematical overview). Criteria for selecting the right AI tool for a given task — accuracy needs, cost, data sensitivity, and output format. Setting up accounts and workspaces; understanding free-tier versus paid-tier capabilities and limitations.

Widely Used AI Tools by Application (for classroom coverage):

Category	Example Tools	Typical Application
Conversational/General-Purpose AI	ChatGPT, Claude, Gemini, Perplexity	Writing, research, coding help, file/data analysis, cited web search
AI Coding Assistants	GitHub Copilot, Cursor, Claude Code	In - IDE code completion, AI Native editing, multi-file agentic coding

	Data Analysis Tools	ChatGPT(Advanced Data Analysis), Microsoft Copilotin Excel, Julius AI	Upload a dataset, generate statistics/charts, spreadsheet Formulas via natural language
	Image Generation	Mid journey, DALL·E 3, Stable Diffusion/Flux	Illustrations, concept art, design mockups
	Writing & Productivity	Grammarly, Gamma, Fathom, Notion AI	Grammar/style checking, slide generation, meeting summaries, task automation
	Audio, Video & Presentation (optional)	Eleven Labs, Runway/Sora, Synthesia	Text-to-speech, AI video generation, script-to-video presentations

Teacher Practical Examples (for classroom demonstration):

1. Live comparison of the same general engineering query (e.g., "explain the engineering design process") answered by two different AI chatbots, highlighting differences in response style and accuracy.

2. Walkthrough of an AI tool's interface — settings, conversation history, file upload, and model selection — using a common platform.

Student Exercise Problems (for practice):

1. Create accounts on two AI platforms and compare their responses to a simple engineering question (e.g., "explain the difference between analysis and design").

2. Prepare a one-page comparison chart of five AI tools by category and primary use case.

3. Write a short reflection identifying one task in your daily academic life where an AI tool could help.

2

Prompt Engineering and Generative AI Tools

Anatomy of a good prompt — context, clear instruction, desired format, and constraints.

Zero-shot, few-shot, and chain-of-thought prompting techniques and when to use each.

Role-based and context-setting prompts (e.g., asking the AI to respond as a reviewer, tutor, or domain expert).

Iterative prompt refinement — diagnosing a poor AI response and rewriting the prompt to fix it.

Using generative AI for brainstorming, summarizing, and drafting technical content.

Teacher Practical Examples (for classroom demonstration):

1. Refine a vague prompt (e.g., "explain project scheduling") into a precise one step-by-step, showing output improvement at each stage.

2. Demonstrate few-shot prompting by providing examples to guide the AI toward a specific output format (e.g., a formatted comparison table).

Student Exercise Problems (for practice):

1. Write three versions of a prompt (basic, detailed, role-based) for the same task and compare outputs.

2. Use an AI tool to summarize a one-page technical article and verify the summary against the original.

3. Draft a chain-of-thought prompt to solve a multi-step numerical/logical problem (e.g., a unit-

	conversion or estimation problem) and check the reasoning.
3	**AI Tools for Data Analysis and Visualization** AI-assisted spreadsheet and dataset analysis using natural-language commands. Natural-language querying of datasets to extract summary statistics and insights. AI tools for generating charts, graphs, and visual summaries from tabular data. Interpreting AI-generated statistical summaries correctly and recognizing their limitations. Identifying and correcting errors or misinterpretations in AI-generated data analysis. **Teacher Practical Examples (for classroom demonstration):** **1.** Upload a small dataset (e.g., class attendance or survey data) to an AI tool and generate a summary-statistics report and chart using natural-language commands. **2.** Show an example where an AI tool misinterprets data (e.g., treats a numeric code as text), and demonstrate how to verify and correct it. **Student Exercise Problems (for practice):** **1.** Use an AI tool to clean and summarize a small sample dataset (e.g., class attendance or marks data). **2.** Generate three different chart types from the same dataset using AI prompts and justify which is most appropriate. **3.** Cross-check one AI-generated statistical claim manually and report any discrepancy found.
4	**AI Tools for Coding, Writing, and Productivity** AI coding assistants — code generation, explanation, and debugging support for simple programs. AI writing assistants for reports, emails, and technical documentation. AI tools for scheduling, note-taking, and meeting/lecture summarization. AI-assisted presentation and slide-outline generation from a topic prompt. Integrating AI tools into an efficient day-to-day academic and engineering workflow. **Teacher Practical Examples (for classroom demonstration):** **1.** Use an AI coding assistant to generate a simple program (e.g., a temperature converter), then intentionally introduce a bug and have the AI debug it. **2.** Generate a presentation outline and speaker notes from a topic prompt using an AI writing/productivity tool. **Student Exercise Problems (for practice):** **1.** Use an AI coding tool to write and debug a simple program for a given problem statement. **2.** Draft a formal email or short report using an AI writing assistant, then manually edit it for accuracy and tone. **3.** Generate a slide outline for a 5-minute seminar topic using an AI tool and refine it.
5	**Ethics, Reliability, and Emerging Trends** Bias and hallucination in AI outputs, and techniques for detecting and fact-checking them. Academic integrity and appropriate citation of AI assistance in coursework and submissions. Data privacy and security considerations, including obligations under India's Digital Personal Data Protection Act, 2023 (DPDP Act), when uploading personal or institutional data. Copyright and intellectual property considerations in AI-generated text and image content. Emerging trends — multimodal AI and AI agents — and their implications for future

	engineering practice. **Teacher Practical Examples (for classroom demonstration):** **1.** Show a documented case of AI hallucination (a fabricated fact or citation) and demonstrate how to fact-check it. **2.** Walk through an example of properly citing AI assistance in an academic submission per institutional guidelines. **3.** Walk through a worked example of checking whether AI-generated text or image output raises a copyright/IP concern (e.g., close paraphrasing of a source). **Student Exercise Problems (for practice):** **1.** Identify and document a hallucination or bias instance produced by an AI tool on a factual query. **2**. Draft an "AI usage declaration" for a hypothetical assignment, citing tools and extent of use appropriately. **3.** Review the data-handling/privacy policy of one AI tool used in this course and identify one DPDP Act, 2023 compliance consideration relevant to student data.
Text Books:	
1	Marr, B. (2024). Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society. Wiley.
2	Mollick, E. (2024). Co-intelligence: Living and working with AI. Portfolio/Penguin.
Reference Books:	
1	Selwyn, N. (2022). Education and technology: Key issues and debates (3rd ed.). Bloomsbury.
2	Susskind, R., & Susskind, D. (2022). The future of the professions: How technology will transform the work of human experts (Updated ed.). Oxford University Press.
3	Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.
Web Resources:	
1	Google. (n.d.). Google AI essentials. Grow with Google. https://grow.google/ai-essentials/
2	OpenAI. (n.d.). OpenAI documentation and guides. https://platform.openai.com/docs
3	Anthropic. (n.d.). Prompt engineering overview. Claude Docs. https://docs.claude.com/en/docs/build-with-claude/prompt- Engineering/overview
4	UNESCO. (2023). Guidance for generative AI in education and research. https://unesdoc.unesco.org/
5	AICTE. (2025). AICTE inputs on Artificial Intelligence — Year of AI initiative. https://www.aicte.gov.in/downloads/initiatives/AICTE%20inputs%20on%20Artificial%20Intelligence%20-compre hensive.pdf
6	Ministry of Electronics and Information Technology, Government of India. (2023). Digital Personal Data Protection Act, 2023. https://www.meity.gov.in/content/digital-personal-data-protection-act-2023

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610203	BS	--	--	3	1.5	50	50	3 Hrs.

PHYSICS FOR ELECTRICAL SYSTEMS LAB

(For EEE)

Course Objectives:

1	To impart hands on experience to the students entering Engineering & Technology education about handling equipment/instruments and use them in experimentation.
2	To make the students understand theoretical aspects of various phenomena experimentally.

Course Outcomes:

S.No.	Outcome	Knowledge Level
1	Apply fundamental principles of Physics to conduct engineering laboratory experiments.	K3
2	Get hands on experience in setting up experiments and using the instruments/equipment individually and conduct experiments.	K3
3	Apply graphical and analytical methods to analyze experimental data and determine physical parameters.	K3
4	Get introduced to using new/advanced technologies and understand their significance .	K3

List of Experiments

(Any **10** of the following listed experiments)

(Out of which any **TWO** experiments may be conducted in virtual mode)

1	Determination of thickness of thin object by wedge method.
2	Determination of radius of curvature of a given plano convex lens by forming Newton's rings.
3	Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
4	Determination of resolving power of telescope.
5	Determination of dielectric constant of given capacitor by charging and discharging method.
6	Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
7	Determination of temperature coefficients of a thermistor.
8	Determination of wavelength of Laser and grating element.
9	Estimation of Planck's constant using photoelectric effect.
10	Determination of the Resistivity of semiconductor by four probe method.
11	Determination of the energy gap of a semiconductor using p-n junction diode.
12	Study the variation of magnetic field along the axis of a current carrying circular coil by Stewart & Gee's Method
13	Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall Effect.
14	Determination of Acceleration due to gravity and Radius of gyration using Compound pendulum.

15	Determination of magnetic susceptibility by Kundt's tube method
16	Verification of Brewster's law.
17	Determination of frequency of A.C. supply using Sonometer.
18	Determination of frequency of electrically maintained tuning fork by Melde's apparatus.
19	Determination of Rigidity modulus of material of a wire-dynamic method (Torsional pendulum).
Reference Books:	
1	S. Balasubramanian, M.N. Srinivasan "A Text Book of Practical Physics"- S Chand Publishers, 2017.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610015	ES	--	--	3	1.5	50	50	3 Hrs

IT WORKSHOP

(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, EEE, IT)

Course Objectives:

1.	Understand the basic components of a computer, assemble and disassemble a PC, and install an operating system.
2.	Enable students to connect to and navigate the Internet safely, and to understand netiquette and cyber hygiene practices.
3.	Develop proficiency in office automation tools (Word, Excel, and PowerPoint) for academic and professional documentation, data analysis, and presentation.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	Identify computer hardware components and assemble, disassemble and set up a working PC with the required operating system.	K3
2.	Apply basic network, browser, search, and cyber security skills to access the Internet, troubleshoot connectivity problems, and practice safe online communication.	K3
3.	Build formatted technical documents such as certificates, abstracts and newsletters using Word	K3
4.	Test for formulas, functions and charts on spreadsheets for data analysis, and design structured interactive presentations.	K4

SYLLABUS

1	**Identifying Components:** Identify the peripherals of a computer and the components inside the CPU cabinet (motherboard, processor, RAM, SMPS, storage drives such as HDD/SSD/NVMe, and graphics card) and their functions. Draw the block diagram of the CPU with the configuration of each component.
2	**Assembling a PC** : Disassemble a PC and reassemble it back to working condition. The lab instructor verifies the work and conducts a viva; refer to the demonstration video provided with the course content.
3	**Operating System Installation:** Individually install Windows and Linux on a personal computer. The lab instructor verifies the installation and conducts a viva.
4	**Network Connectivity :** Connect the system to a LAN/Wi-Fi and access the Internet by configuring TCP/IP settings (IP address, DNS). Demonstrate access to websites and email, and use basic troubleshooting commands (ping, ipconfig/ifconfig, tracert).

5	**Browsers, Search and Netiquette:** Customize a web browser with bookmarks, pop-up blockers and useful extensions (PDF viewer, ad blocker, password manager). Use search engines effectively with filters and Boolean operators, and demonstrate proper netiquette for online communication.
6	**Cyber Hygiene :** Identify common online threats (phishing, malware, unsafe downloads) and configure the system for safe browsing: HTTPS-only browsing, strong passwords with a password manager, two-factor authentication, and keeping the OS and antivirus updated.
7	**Certificate Creation :** Using MS Word , Create a project certificate covering fonts, drop cap, text effects, character spacing, borders and colours, headers and footers, and date/time options.
8	**Project Abstract:** Create a project document on a topic of your choice, and demonstrate the use of formatting styles, tables, bullets and numbering, text direction, cell alignment, footnotes, hyperlinks, symbols, spell check, and track changes.
9	**Newsletter, and Introduction :** Create a newsletter covering table of contents, newspaper columns, images and clip art, the drawing toolbar and WordArt, image formatting, text boxes, paragraph formatting and mail merge.
10	**Spreadsheet Scheduler and GPA:** Using MS Excel, create a scheduler (gridlines, cell formatting, AutoFill, summation) and a GPA sheet using cell referencing, the AVERAGE, STDEV and COUNT functions, charts, and multiple worksheets.
11	**Spreadsheet Data Analysis:** Perform data analysis using LOOKUP/VLOOKUP, splitting cells, freezing panes, grouping/outlining, sorting, Boolean and logical operators, conditional formatting and a basic pivot table.
12	**Interactive Presentation:** Create an interactive presentation covering slide layouts, WordArt, bullets and autoshapes, hyperlinks, inserting images, audio, video, tables and charts; master layouts (slide, template, notes),views and backgrounds and design templates.
Reference Books:	
1	Computer Fundamentals, Anita Goel, Pearson India Education,2017
2	P. K. Sinha and Priti Sinha, Computer Fundamentals, BPB Publications.
3.	Rajaraman V., Fundamentals of Computers, PHI Learning.
4.	Joan Lambert, Microsoft Office 2019/2021 Step by Step, Microsoft Press.
Web References:	
1.	Virtual Labs – Computer Organisation and Hardware: https://www.vlab.co.in/broad-area-computerscience-and-engineering
2.	AWS Free Tier: https://aws.amazon.com/free/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610204	ES	--	--	3	1.5	50	50	3 Hrs

BASIC ELECTRICAL CIRCUITS LABORATORY

(For EEE)

Course Objectives: Student will learn

1	About the basic principles of electrical circuits and wiring.
2	To verify circuit laws and theorems.
3	About safety measures used in electrical systems.
4	About measurement of electrical quantities, parameters of choke coil and magnetically coupled circuits.
5	About the phasor diagram of RLC Circuit.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1	Implement domestic wiring for staircase and energy measurement experimentally.	K3
2	Verify Ohm's Law, Kirchhoff's Laws and Superposition Theorem experimentally using DC circuits.	K3
3	Determine the parameters of choke coil, magnetically coupled coils and power in AC circuits.	K3
4	Test the operation of safety equipment under different fault conditions for safety awareness.	K4
5	Illustrate RLC phasor diagram experimentally and measure amplitude & frequency of waveforms using CRO	K3

List of Experiments

	Demonstration of Electrical tools, wiring accessories, Electrical cables/conductors, switches, sockets, Moving Iron meters, Moving Coil meters, Digital multi-meter, fuses.
1	Two-Way Switching of a Lamp Using Staircase Wiring
2	Study of Domestic Wiring and Measurement of Electrical Energy for different loads
3	Verification of Kirchhoff's laws
4	Verification of Superposition Theorem
5	Verification of Ohm's Law & filament lamp
6	Overload and Short circuit protection using Fuse / Miniature Circuit Breaker (MCB).
7	Determination of parameters of a choke coil
8	Determination of Coefficient of Coupling in Magnetically Coupled Coils
9	Measurement of Earth resistance and application of ELCB / RCCB.
10	Measurement of Power in 1-Phase AC Circuit with Fan Load.

11	Phasor Diagram of a Series RLC Circuit.
12	Measurement of amplitude and frequency of different voltage waveforms using CRO.
Reference Books:	
1	Principles of Electrical Engineering, V.K Mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2017.
2	S. L. Uppal & G. C. Garg, (2018), Electrical Wiring, Estimating and Costing, 6th edition, Khanna Publishers.
3	Network Analysis, M. E. Van Valkenburg, Pearson Education, 2019, Revised 3rd Edition.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610018	MC	--	--	2	--	50	--	--

NSS, SPORTS & YOGA

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

Course Objectives:

1.	The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.
2.	To make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand the importance of discipline, character and service motto.	K2
2.	Explore human relationships by analyzing social problems.	K3
3.	Develop leadership skills and civic responsibilities.	K3
4.	Understand the importance of yoga and sports for physical fitness and sound health	K2
5.	Demonstrate health related fitness components.	K2

SYLLABUS

UNIT-I (10Hrs)	**Orientation** : General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance. **Activities:** i Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills ii Conducting orientations programs for the students –future plans-activities-releasing road map etc. iii Displaying success stories-motivational biopics- award winning movies on societal issues etc. iv Conducting talent show in singing patriotic songs-paintings- any other contribution.
UNIT-II (10 Hrs)	**Nature & Care** **Activities:** i Best out of waste competition. ii Poster and signs making competition to spread environmental awareness. iii Recycling and environmental pollution article writing competition. iv Organising Zero-waste day. v Digital Environmental awareness activity via various social media platforms. vi Virtual demonstration of different eco-friendly approaches for sustainable living.

	vii Write a summary on any book related to environmental issues.
UNIT-III (10 Hrs)	**Community Service** **Activities:** i Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc. i Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS, ii Conducting consumer Awareness. Explaining various legal provisions etc. iii Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education. iv Any other programmes in collaboration with local charities, NGOs etc.
UNIT-IV (10 Hrs)	Yoga Activities: Yoga practices- Asana, Surya Namaskar, Meditation.
UNIT-V (10 Hrs)	Physical Activities: Practicing general and specific warm up, Aerobics. Cardio respiratory fitness, Skipping and Running. Practice one major game and one individual sport(volley ball, basket ball).

Reference Books:	
1.	Nirmalya Kumar Sinha & Surajit Majumder, A Text Book of National Service Scheme Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2.	Red Book-National Cadet Corps-Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3.	Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008
4.	Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007
5.	Ram Ahuja. Social Problems in India, Rawat Publications, New Delhi.
6.	The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014
e-Resources	
1.	https://books.google.co.in/books/about/Social_Problems_in_India.html?id=jsvJoAEACAAJ&redir_esc=y
2.	https://openlibrary.org/verify_human?next=/books/OL32142863M/Social_problems_in_India%3Futm_source%3Dchatgpt.com

Regulation:R26	I/IV-B.Tech. II -Semester

ELECTRICAL & ELECTRONICS ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2620001	Differential Equations and Vector calculus	**BS**	3	0	0	3	40	60	100
A2620201	Chemistry for Electrical Systems	**BS**	3	0	0	3	40	60	100
A2620202	Electrical Network Analysis	**PC**	3	0	0	3	40	60	100
A2620203	Computational Thinking and Problem Solving with C	**ES**	3	0	0	3	40	60	100
A2620009	Engineering Graphics	**ES**	0	1	3	2.5	40	60	100
A2620204	Networks Laboratory	**PC**	0	0	3	1.5	50	50	100
A2620205	Computational Thinking and Problem Solving using C lab	**ES**	0	0	3	1.5	50	50	100
A2620206	Chemistry for Electrical Systems lab	**BS**	0	0	3	1.5	50	50	100
A2620019	English for Technical Communication and MOODLES Lab	**HS**	0	0	3	1.5	50	50	100
A2620020	Universal Human Values - II	**MC**	2	0	0	0	50	-	50
		TOTAL	**14**	**1**	**15**	**20.5**	**450**	**500**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620001	BS	3	--	--	3	40	60	3 Hrs.

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common for all Branches)

Course Objectives: Students are expected to learn

1.	First order ordinary differential equations and some simple geometrical and physical applications.
2.	Methods of solution of linear higher order ordinary differential equations.
3.	Formation and solution of linear partial differential equations.
4.	Concepts of Gradient, divergence, curl.
5.	Vector integral theorems.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.	K3
2.	**Solve** linear ordinary differential equations of second and higher order which occur in various engineering fields.	K3
3.	**Apply** the appropriate methods of solution for partial differential equations that model physical processes.	K3
4.	**Compute** the gradient of a scalar field, curl and divergence of various vector fields to demonstrate their physical behavior.	K3
5.	**Calculate** the work done against a field, circulation and flux using vector calculus.	K3

SYLLABUS

UNIT-I (10Hrs)	**Differential equations of first order and first degree** Exact equations and equations reducible to exact form (Integrating Factor), Linear differential equations, Bernoulli's equations. Orthogonal Trajectories, Applications: Newton's Law of cooling, Law of natural growth and decay, Electrical circuits: RL & RC circuits.
UNIT-II (10 Hrs)	**Linear differential equations of higher order (Constant Coefficients)** Definitions, homogenous and non-homogenous, complimentary function, particular integral, general solution, Method of variation of parameters – Wronskian. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.
UNIT-III (10 Hrs)	**Partial Differential Equations** Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous and Non-Homogeneous Linear Partial differential equations with

	constant coefficients.
UNIT-IV (10 Hrs)	**Vector Differentiation** Scalar and vector point functions, vector operator Del, Gradient and applications, Directional derivative, Divergence – Solenoidal and Curl – Irrotational Field – Scalar Potential, Vectoridentities.
UNIT-V (10 Hrs)	**Vector Integration** Line Integral – Circulation – Work done by force, surface integral – flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and their applications.

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition.
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2.	Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5.	Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty two Edition (Reprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.

e-Resources	
1.	https://mathworld.wolfram.com/First-OrderOrdinaryDifferentialEquation.html
2.	https://en.wikipedia.org/wiki/Differential_equation
3.	https://en.wikipedia.org/wiki/Partial_differential_equation
4.	https://en.wikipedia.org/wiki/Vector_calculus
5.	https://en.wikipedia.org/wiki/Vector_calculus

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620201	BS	3	--	--	3	40	60	100

CHEMISTRY FOR ELECTRICAL SYSTEMS

(For EEE)

Course Objectives:

1.	To develop an understanding of electrochemical principles, corrosion mechanisms, and corrosion protection techniques relevant to electrical and electronic systems.
2.	To develop knowledge of chemical fuels, electrochemical energy-storage devices, fuel cells.
3.	To develop an understanding of the chemistry, properties, and applications of semiconductors, polymers, and nanomaterials used in electrical and electronic systems.
4.	To develop knowledge of the principles and applications of electrochemical sensors, biosensors, colorimetry, UV-Visible spectroscopy, and IR spectroscopy for material and chemical characterization.
5.	To develop an understanding of quantum chemistry and molecular orbital concepts relevant to electronic structure, bonding, and advanced electronic materials.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	**Apply** electrochemical principles to **analyze** corrosion mechanisms and evaluate appropriate corrosion protection techniques for electrical and electronic materials and systems.	K3, K4
2.	**Apply** electrochemical principles to **analyze** the characteristics, working mechanisms, and applications of batteries, fuel cells.	K3, K4
3.	**Apply** materials chemistry principles to **analyze** the properties and applications of semiconductor materials, polymers, and nanomaterials used in electrical and electronic systems.	K3, K4
4.	**Apply** the principles of sensors, biosensors, colorimetry, UV-Visible spectroscopy, and IR spectroscopy to **analyze** chemical species and characterize materials relevant to electrical and electronic applications.	K3, K4
5.	**Apply** quantum chemistry principles to **analyze** electronic structure, molecular orbitals, bonding, and energy levels relevant to electrical and electronic materials.	K3, K4

SYLLABUS

UNIT-I (10Hrs)	**Electrochemistry and Corrosion** • Fundamentals of electrochemistry: Electrochemical cells, Electrode potential, Nernst equation, , Reference electrodes -Calomel electrode, (Construction and working), Electro-chemical series and its uses. Conductometric titrations: strong acid vs strong base titrations. • Corrosion: Definition, chemical and electro-chemical corrosion; Pilling-Bed worth rule, Galvanic corrosion, differential aeration corrosion, pitting, stress corrosion,

	factors affecting corrosion. • Protective techniques: Material selection and proper design, cathodic protection (sacrificial anodic and impressed-current protection). Galvanizing, electroplating and electroless plating (nickel and copper). Corrosion penetration rate (CPR).
UNIT-II (10 Hrs)	**Energy Conversion and Storage** • Chemical Fuels: characteristics of a fuel, calorific value, determination of calorific value using bomb calorimeter, numerical problems on GCV , Green fuels: Synthesis of Green hydrogen, hythane. • Batteries: Introduction, classification of batteries, characteristics of battery-capacity, power density, cell balancing and cycle life. • Primary batteries: construction, working and applications of zinc-air and Lithium batteries • Secondary batteries: lead-acid, Lithium-ion batteries — construction, charging-discharging reactions and applications. • Fuel cells: Hydrogen-oxygen, Polymer electrolyte fuel cells (PEM) and solid-oxide fuel cell (SOFCs) applications and limitations.
UNIT-III (10 Hrs)	**Electrical & Electronic Materials** • Intrinsic and extrinsic semiconductors, Preparation of pure semiconductor material (Zone refining), Chemistry of compound semiconductors (GaAs, InP, SiC), Advanced semiconducting materials-Organic(Introduction, applications,Conduction mechanism of Polyacetylene) and Perovskite semiconductors (Definition,applications) • Polymers: Thermoplastics and thermosetting plastics — cross-linked polyethylene (XLPE), PVC, PTFE, PEEK, polyimide, epoxy resin, silicone rubber (preparation, properties, applications). • Nanomaterials: CNT, graphene - preparation (CVD), properties and electronic applications
UNIT-IV (10 Hrs)	**Spectroscopy-Instrumental methods and applications** Electromagnetic spectrum, • UV-Visible spectroscopy: Beer-Lambert law, principle, Electronic transitions instrumentation, chromophore, auxochrome, bathochromic shift, hypsochromic shift, applications. • Infrared (IR) spectroscopy: Introduction, principle, Fundamental modes of vibrations, instrumentation and applications. • Microwave spectroscopy: Introduction, principle, instrumentation and applications. • MRI: Working principle, applications.

UNIT-V (10 Hrs)	**Quantum Chemistry** • Fundamentals of quantum chemistry: Black-body radiation, photoelectric effect, wave-particle duality, de Broglie wave equation, Heisenberg's uncertainty principle. • Fundamentals of quantum mechanics: Schrödinger wave equation (time independent), significance of ΨandΨ^2; particle in a one-dimensional box. • Molecular orbital theory: bonding in homo-and hetero-nuclear di atomic molecules; energy-level diagrams of O2, N2, HF and CO: calculation of bond order • π-molecular orbitals of butadiene and benzene;

Textbooks:	
1.	Agarwal, S. (2026). Engineering Chemistry (5th ed.). Cambridge University Press.
2.	Jain, P. C., & Jain, M. (2020). Engineering Chemistry (17th ed.). Dhanpat Rai Publishing Company
Reference Books:	
1.	Linden, D., & Reddy, T. B. (Eds.). (2001). Handbook of batteries (3rd ed.). McGraw-Hill.
2.	Roberge, P. R. (2008). Corrosion engineering: Principles and practice. McGraw-Hill.
3.	Shikha, A., & Sharma, R. K. (2019). Engineering Chemistry. Krishna Prakashan Media
e-Resources	
1.	IEEE Standards Association. (n.d.). IEEE standards. https://standards.ieee.org
2.	International Electrotechnical Commission. (n.d.). IEC standards for electrical equipment. https://www.iec.ch
3	International Renewable Energy Agency. (n.d.). Renewable energy technologies. https://www.irena.org
4	U.S. Department of Energy. (n.d.). Energy storage database. https://www.energy.gov

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620202	PC	3	--	--	3	40	60	3 Hrs

ELECTRICAL NETWORK ANALYSIS

(For EEE)

Course Objectives: Student will learn about

1.	The transient behavior of first-order DC circuits using differential equations.
2.	The transient behavior of second-order DC circuits using differential equations.
3.	The application of Laplace transforms to electrical circuits.
4.	The concept of two-port networks and network functions.
5.	The concepts of balanced and unbalanced three-phase circuits.

Course Outcomes: At the end of this course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Analyze DC transients in first order RL & RC circuits using classical methods for solving differential equations.	**K4**
2.	Analyze DC transients in second order RLC circuits using classical methods for solving differential equations.	**K4**
3.	Analyze RLC circuit transients using Laplace transforms with DC and AC excitation.	**K4**
4.	Determine two-port network parameters and network functions for RLC networks.	**K4**
5.	Analyze the balanced and unbalanced three-phase circuits.	**K4**

SYLLABUS

UNIT- I (10 Hrs)	**DC Transient Analysis-First order circuits:** Inductance and Capacitance transient behavior, Introduction to first order circuits, Source-free RC circuit, Source-free RL circuit, Evaluation of initial conditions, Step Response of an RC circuit, Step Response of an RL circuit, Concepts of Natural, Forced and Complete Response.
UNIT-II (10 Hrs)	**DC Transient Analysis-Second order circuits:** Introduction to Second order circuits, Source- free series RLC circuit, Source- free Parallel RLC circuit, Step Response of a series RLC circuit, Step Response of a parallel RLC circuit, The lossless LC circuit.
UNIT-III (10 Hrs)	**Laplace Transform Techniques for Transient Analysis:** The Laplace transformation, Definition and Basic theorems of Laplace transformation, Initial and Final value theorems, Inverse Laplace transforms, Laplace transform of typical signals, Laplace transformation application to simple circuits considering unit step and sinusoidal forcing functions.

UNIT-IV (10 Hrs)	**Two Port Networks and Network Functions** Two port network parameters**:** Relationship of Two-Port Variables, Short circuit Admittance (Y)-Parameters, Open circuit Impedance (Z)-Parameters, Hybrid (H)-Parameters, Transmission (ABCD) Parameters, two port parameters for resistive and simple AC Networks, Relationships between Parameter sets, Series, Parallel and Cascade connection of two port networks. Network Functions: The concept of Complex frequency, transform impedance and transform circuits, Network functions for the one port and two ports.
UNIT-V (10 Hrs)	**Three phase AC circuits:** Advantages of Three Phase system, Phase sequence, Relation between Line and Phase Quantities in Star and delta connected circuits, Power in Three-phase circuits, Analysis of Balanced & Unbalanced Three Phase Circuits, Three-Phase Power Measurements (Two-Wattmeter method).
Textbooks:	
1.	Hayt, W. H., & Kemmerly, J. E, (2018). Engineering circuit analysis (9th edition.). McGraw Hill Education.
2.	Van Valkenburg, M. E, (2019), Network analysis (3rd edition.). Prentice-Hall of India.
Reference Books:	
1.	Alexander, C. K., & Sadiku, M. N. O. (2019). Fundamentals of electric circuits (6th edition.). McGraw Hill Education.
2.	Schaum's Outlines Electric Circuits,Nahvi, Mahmood, Ph.D., Edminister, Joseph A,7th Edition, Mcgraw Hill.
3.	Network Analysis and Synthesis, by Ravish R. Singh, McGraw Hill Education (1 July 2017).
4.	Network Analysis, 3rd Ed, A Sudhakar Shyammohan.SPalli Tata McGraw Hill Education Pvt Ltd.
5.	Circuit Theory Analysis and Synthesis.,7th Edition 2014 by Abhijit Chakrabarthi, DhanpatRai &Co.
e-Resources	
1.	nptel.ac.in/courses/117106108
2.	nptel.ac.in/courses/108105159

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620203	ES	3	--	--	3	40	60	3 Hrs

COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C

(For EEE)

Pre-requisite: Mathematics and logical reasoning at 10+2 level. No prior programming experience is assumed.

Course Objectives: The main objectives of the course are to

1.	Familiarize students with computers, computational thinking, and the process of algorithmic problem solving.
2.	Build strong programming fundamentals using the C language, including operators and control structures.
3.	Develop the ability to design and implement solutions using arrays, strings, and user- defined data types.
4.	Impart working knowledge of pointers, recursion, dynamic memory allocation, functions, and file handling for building modular, reusable programs.

Course Outcomes: At the end of the course, student will be able to

S.No	Outcome	Knowledge Level
1.	Apply computational thinking strategies to construct algorithms, flowcharts, and implement them as C programs.	K3
2.	Apply selection and iterative control structures along with arrays to solve programming problems.	K3
3.	Analyze problems involving strings and user-defined datatypes to construct suitable C solutions.	K4
4.	Apply pointers, dynamic memory allocation, and recursive techniques to design modular C programs using functions.	K3
5.	Apply preprocessor directives and file handling techniques to perform file input-output operations in C programs.	K3

SYLLABUS

UNIT-I (10Hrs)	**Computational Thinking and Introduction to C** **Computational Thinking:** Decomposition, Pattern Recognition, Abstraction, Algorithm Design; representing solutions as algorithms, pseudocode, and flowcharts; Number Systems and Data Representation (integers, reals); Overview of Computer Organization and Program Execution. **Introduction to C:** Background, Structure of a C Program, Identifiers, Keywords, Variables, Constants, Basic Data Types, Input/Output; Expressions, Operators, type of operators- arithmetic, logical, relational, assignment, conditional, increment,

	decrement, etc., unary, binary, ternary, Operator Precedence and Associativity, Type Conversion; Scope, Storage Classes, and Type Qualifiers. **Bitwise Operators:** Logical Bitwise Operators, Shift Operators.
UNIT-II (10 Hrs)	**Selection and Decision-Making**: Two-Way and Multi-Way Selection (if-else, switch), Nested Conditionals. **Repetition:** Pre-test and Post-test Loops, Counter- and Event-Controlled Loops, for/while/do-while, break and continue, Nested Loops; Programming Applications combining Selection and Repetition. **Arrays:** One-Dimensional and Two-Dimensional Arrays, Multidimensional Arrays, Array-based Problem Solving (searching, sorting, averaging).
UNIT-III (10 Hrs)	**Strings:** String Representation, Standard String-Handling Functions, Arrays of Strings, String-to-Data Conversion. **User-Defined Types:** typedef, Enumerated Types, Structures, Unions, and Nested Structures; Programming Applications using Structures and Arrays of Structures
UNIT-IV (10 Hrs)	**Pointers:** Pointer Fundamentals, Pointer-to-Pointer, Pointer Arithmetic, Pointers and Arrays, Array of Pointers, Compatibility and l-value/r-value concepts. Dynamic Memory Allocation: malloc, calloc, realloc, free. **Functions:** Designing Structured/Modular Programs, Function Declaration and Definition, Parameter Passing (by value, by reference using pointers), Passing Arrays and Pointers to Functions, Standard Library Functions, Inter-Function Communication. **Recursion:** Recursive vs. Iterative Solutions, Designing Recursive Algorithms, Recursion with Arrays and Pointers, Efficiency Considerations.
UNIT-V (10 Hrs)	**Preprocessor**: Directives and Macros. **File I/O:** Streams, Text vs. Binary Files, Standard Library File Functions, formatted and Character I/O, Sequential File Processing.
Textbooks:	
1.	Behrouz A. Forouzan & Richard F. Gilberg, Computer Science: A Structured Programming Approach Using C, 3rd Edition, Cengage Learning, ISBN-13: 978-0-534-49132-1.
2.	Brian W. Kernighan & Dennis M. Ritchie, The C Programming Language, 2nd Edition, Pearson Education / Prentice Hall India, 1988.
Reference Books:	
1.	Sumitabha Das, "Computer Fundamentals and Programming", 18th Edition, McGraw Hill Education, 2018
2.	Ashok N. Kamthane, Amit Ashok Kamthane, "Programming in C", 4th Edition, Pearson Education, 2025. ISBN: 978-9332543633.
3.	Pradip Dey, Manas Ghosh, "Computer Fundamentals and Programming in C", 2nd Edition, Oxford University Press, 2013. ISBN: 978-0198068842

e-Resources	
1.	NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview
2.	Geeks for Geeks – C Programming Language: https://www.geeksforgeeks.org/c-programming-language/
3.	Programiz – Learn C Programming: https://www.programiz.com/c-programming
4.	cplusplus.com – C Standard Library Reference: https://cplusplus.com/reference/clibrary/
5.	TutorialsPoint – C Programming: https://www.tutorialspoint.com/cprogramming/index.html
6.	W3Schools – C Tutorial: https://www.w3schools.com/c/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620009	ES	--	1	3	2.5	50	50	3 Hrs

ENGINEERING GRAPHICS

(Common to CE, EEE & ME)

Course Objectives:

1	Develop spatial visualization and graphical communication skills to represent three-dimensional engineering objects using standard engineering graphics principles.
2	Create 2D, 3D CAD models and communicate engineering design information effectively through conventional drawings and digital CAD tools.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	**Apply** standard engineering graphics principles and conventions to prepare accurate sketches and engineering drawings.	**K3**
2	**Visualize** and represent three-dimensional objects and systems using appropriate projection, orthographic, isometric, sectional, and surface-development techniques.	**K4**
3	Use CAD tools to **Create**, modify, and **Communicate** 2D drawings and 3D models with appropriate dimensions, annotations, symbols, and features.	**K4**
4	**Generate** complete engineering design documentation through original graphical representations, detailed CAD models, and compliant engineering drawings for multidisciplinary applications.	**K4**

SYLLABUS

Section A – Manual Engineering Drawing

1. **Geometrical Constructions and Engineering Curves:**
   Introduction to Engineering Drawing, Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.
2. **Engineering Curves**: Parabola, Ellipse and Hyperbola by general method (Eccentricity method only), Cycloidal curves, Involutes, tangent & normal for these curves.
3. **Projection of Points and Straight Lines:** Projection of points in different quadrants and straight lines in different orientations.
4. **Projections of planes:** projections of regular planes perpendicular to one reference plane and parallel or inclined to the other, Planes inclined to both reference planes.
5. **Projections of Solids**: Types of solids- Polyhedra and Solids of revolution. Projections of solids in simple positions.
6. **Sections of Solids**: Sections and Sectional views of Right and Regular Solids.
7. **Isometric Projection:** Introduction to Isometric projection and Isometric projection of simple Right and Regular Solids – Prism, Cylinder, Pyramid and Cone.

Section B –Computer Aided Engineering Graphics

1. **Computer-Aided Engineering Graphics (CAD)** Introduction to CAD software, standard tool bar menu and description of most commonly used tool bars, and navigational tools.
2. **CAD Geometrical Constructions and Curves**: Construction of polygons, circles, arcs, ellipse and other engineering curves using CAD.
3. **2D CAD – Orthographic Views**: Preparation of front, top, side views of simple engineering objects
4. **Basic 3D CAD Modelling:** Creation of simple three-dimensional engineering objects using basic modelling features.
5. **3D CAD Modelling with Features:** Creation and modification of 3D models using features such as hole, fillet, chamfer, mirror, pattern and other appropriate tools
6. **Construct a 3D CAD model** of a given engineering object from its front, top and side orthographic views.
7. **CAD Sectional Views:** Creation of sectional views of 3D engineering objects and generation of corresponding 2D drawings.
8. **CAD Surface Development**: Development of cylindrical, conical, prismatic surfaces using CAD.
9. **3D Model to Engineering Drawing:** Generation of orthographic, isometric and sectional views with dimensions, annotations and symbols from a 3D CAD model.
10. **CAD Modelling of Physical Engineering Objects:** Measurement and interpretation of a physical engineering object, recreation of its geometry in CAD, and preparation of its engineering drawing.
Reference Books:

1	Bhatt, N. D., & Panchal, V. M. (2023). Elementary engineering drawing (53rd ed.). Charotar Publishing House.
2	Narayana, K. L., Kannaiah, P., & Reddy, K. V. (2020). Engineering graphics (2nd ed.). Scitech Publications
3	Venugopal, K., & Prabhu Raja, V. (2019). Engineering drawing and graphics (5th ed.). New Age International.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620204	PC	--	--	3	1.5	50	50	3 Hrs.

NETWORKS LABORATORY

(For EEE)

Course Objectives: Students will learn about

1	The electrical network theorems and their verification through laboratory experiments.
2	The series resonant circuit and current locus diagram of series RL & RC circuits.
3	The transient responses of RC & RLC circuits.
4	The practical skills in measuring voltage, power and power factor in three-phase circuits.
5	The determination and application of two-port network parameters such as Z, Y, H and ABCD parameters.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	Verify Thevenin's, Norton's, Reciprocity and Maximum Power Transfer theorems experimentally using DC networks.	K3
2	Analyze series resonant circuit and obtain locus diagrams for series RL & RC circuits.	K4
3	Analyze transient responses of RC & RLC circuits.	K4
4	Determine phase voltage, line voltage, power and power factor in balanced and unbalanced three-phase circuits.	K3
5	Determine Z, Y, H and ABCD parameters of two-port networks.	K3

List of Experiments

1	Verification of Thevenin's theorem.
2	Verification of Norton's theorems
3	Verification of Reciprocity theorem.
4	Verification of maximum power transfer theorem.
5	Series Resonance.
6	Determination of open-circuit (Z), short-circuit (Y) parameters, hybrid (H) and transmission (ABCD) parameters.
7	Locus diagram of Series RL & RC circuit.
8	Measurement of power and power factor in 3-phase unbalanced load by Two-wattmeter method.
9	Measurement of Phase voltage, line voltage and power for balanced 3-phase load.
10	Cascade connection of Two port networks.
11	DC transient response of series RC circuit/ series RLC circuit.
12	Single-Phase UPS/Inverter Circuit.

Reference Books:

1	1. Hayt, W. H., & Kemmerly, J. E. (2018). Engineering circuit analysis (9th ed.). McGraw Hill Education.
2	Alexander, C. K., & Sadiku, M. N. O. (2019). Fundamentals of electric circuits (6th ed.). McGraw Hill Education.
3	Van Valkenburg, M. E. (2019). Network analysis (3rd ed.). Prentice-Hall of India.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620205	ES	--	--	3	1.5	50	50	3 Hrs

COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C LAB

(For EEE)

Pre-requisite: - Basic understanding of computer fundamentals and problem-solving concepts

Course Objectives: The main objectives of the course are to

1	Apply the principles of C language and computational thinking techniques in problem solving.
2	Design flowcharts and algorithms, and to develop the ability to trace and debug programs.
3	Design and develop C programs using arrays, strings, pointers, recursion, and functions.
4	Apply dynamic memory allocation, file operations, and preprocessor commands in program design.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	Apply computational thinking to construct algorithms and flowcharts for a given Problem and translate them into C programs.	K3
2	Develop C programs using operators, selection statements and loops to implement a specified logic.	K3
3	Develop C programs that manipulate arrays, strings, structures and pointers to solve computational problems.	K3
4	Construct modular programs using user-defined functions and recursion and perform file input-output operations.	K3
5	Analyze a C program by tracing and debugging it and evaluate alternative Implementations for correctness and efficiency.	K4

SYLLABUS

1	**Exercise 1: Getting Started with** 1. Write, compile and run a C program that prints "Hello, World!" on the screen. Identify the source file, the object file and the executable produced, and record the command used at each stage. 2. Write a C program that declares variables of type int, float, char and double, reads a value for each from the user, and displays each value together with the size of its type using size of. 3. Write a C program to read the height and width of a rectangle from the user and compute its perimeter and area.

2	**Exercise 2: Operators and Expressions** 1. Write a C program to compute the distance between two points whose coordinates are entered by the user. 2. Write a C program that reads four integers p, q, r, s and, using relational and logical operators, prints 'Correct values' if q > r, s > p and (r + s) > (p + q); otherwise 'Wrong values'. 3. Write a C program to compute simple and compound interest for values entered by the user.
3	**Exercise 3: Selection and Decision Making** 1. Write a C program to read a character and classify it as a vowel, consonant, digit or special symbol. 2. Write a C program to find the largest of three numbers using nested if-else. 3. Write a menu-driven C program to compute the area of various geometrical shapes (circle, rectangle, triangle, square) using switch-case.
4	**Exercise 4: Loops and Series** 1. Write a C program to display the first n even natural numbers and their sum. 2. Write a C program to reverse a given integer and check whether it is a palindrome. 3. Write a C program to display the sum of the harmonic series 1 + 1/2 + 1/3 + ... up to n terms.
5	**Exercise 5: Nested Loops and Number Patterns** 1. Write a C program to print the multiplication table pyramid for numbers 1 to n using nested loops. 2. Write a C program to check whether a given number is prime, then extend it to print all prime numbers and all Armstrong numbers below.
6	**Exercise 6: One-Dimensional Arrays** 1. Write a C program to print all unique elements of an array and implement linear search to find a given element in the array. 2. Write a C program to separate the odd and even elements of an array into two different arrays and implement binary search to find a given element in a sorted array. 3. Write a C program to sort the elements of an array in ascending order using selection sort and compare the efficiency of linear search vs. binary search on the sorted array by counting the number of comparisons each method takes.
7	**Exercise 7: Matrices** . Write a C program to perform addition of two square matrices and display the resultant matrices. . Write a C program to multiply two square matrices and display the resultant matrices. . Write a C program to find the transpose of a given matrix.

8	**Exercise 8: Strings:** Write C programs to perform the following without using string handling functions • **Find the length** of a given string. • **Copy** one string into another. • **Concatenate** two strings. • **Compare** two strings and display whether they are equal, or which one is greater. • **Reverse** a given string (using a separate swap function to exchange characters). • **Search for a character** in a string and display its position (or indicate if not found). • **Count the number of vowels, consonants, and words** in a given string
9	**Exercise 9: Pointers Fundamentals** 1. Write a C program to demonstrate the use of the & (address-of) and * (value-at-address) operators. 2. Write a C program to add two numbers using pointers. 3. Write a C program to swap two values using call by reference.
10	**Exercise 10: Pointers and Dynamic Memory** . Write a C program to dynamically allocate memory for an array using malloc, find the largest element and the sum of all elements in it, then repeat using calloc and note the difference between the two. . Write a C program to count the number of vowels in a string using a pointer.
11	**Exercise 11: User-Defined Functions** . Write a C program to check whether a number is prime using a user-defined function. . Write a C program to compute the GCD and LCM of two numbers using functions. . Write a C program to demonstrate a function that returns a pointer.
12	**Exercise 12: Recursion** . Write a C program to compute the factorial and the n-th Fibonacci term using recursion. . Write a C program to compute the sum of digits of a number using recursion. . Write a C program to convert a decimal number to its binary equivalent using a recursive function.
13	**Exercise 13: File Handling** . Write a C program to append multiple lines at the end of an existing text file. . Write a C program to copy the contents of one file into another file. . Write a C program to count the number of characters, words and lines in a text file.
Reference Books: **Web References:**	
1	NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview
2	Virtual Labs – Problem Solving Lab, IIIT Hyderabad: https://ps-iiith.vlabs.ac.in/
3	GeeksforGeeks – C Programming: https://www.geeksforgeeks.org/c-programming-language/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620206	BS	--	-	3	1.5	50	50	100

CHEMISTRY FOR ELECTRICAL SYSTEMS LAB

(For EEE)

Course Objectives:

1	To develop practical skills in chemical analysis using volumetric, electrochemical, spectroscopic, and instrumental techniques for the characterization of chemical, water, and electronic-grade samples.
2	To develop hands-on skills in emerging chemistry applications through the synthesis and characterization of nanomaterials, polymeric dispersions, and electrochemical biosensors, and to relate experimental observations to technological applications.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	**Apply** volumetric, conductometric, potentiometric, pH-metric, and redox titration techniques to determine the concentration and properties of acids, bases, battery electrolytes, and metal ions.	K3
2	**Apply** analytical and instrumental techniques to determine water-quality parameters, corrosion rate, viscosity, and acid value of materials used in electrical and energy systems.	K3
3	**Apply** spectroscopic, colorimetric, chemical, and electrochemical methods to analyze chemical species, identify functional groups, synthesize polymers and nanomaterials, and investigate electrochemical processes.	K3
4	**Apply** chemical synthesis, material characterization, and computational/virtual-laboratory techniques to study nanomaterials and polymeric materials, hydrogen production by water electrolysis, and electronic energy levels using the particle-in-a-box	K3

SYLLABUS

1	Determination of the concentration/strength of sulfuric acid in Pb-acid battery electrolyte
2	Conductometric titration of a strong acid against a strong base
3	Determination of a strong acid against a strong base by using pH meter.
4	Estimation of Iron in Mohr's salt by potassium dichromate.
5	Determination of a weak acid against a strong base by using potentiometer.
6	Determination of the corrosion rate of a mild-steel specimen by the weight-loss method.
7	Synthesis of bakelite.
8	Synthesis of nanomaterial by chemical precipitation method.
9	Determination of Dissolved oxygen in water samples by Winklers method.
10	To determine the concentration of Fe^{2+}/Fe^{3+} in a given water or leach liquor sample by colorimetry.

11	Determination of the viscosity of an insulating oil using Ostwald viscometre.
12	Determination of the acid value of transformer oil(titration method).
13	Identify functional groups present in a given organic or mineral-bound sample from its IR spectrum. (virtual lab)
14	Study the electrochemical production of hydrogen by water electrolysis. (virtual lab)
15	Calculate and plot allowed energy levels of an electron confined in a one-dimensional box of varying length using the particle-in-a-box model. (virtual Lab)
Reference Books:	
1	Ahluwalia, V. K., & Dhingra, S. (2017). Comprehensive practical organic chemistry. Universities Press
2	Vogel,A.I.(2012), Vogels text book of Quantitative chemical analysis, Pearson Education
3	Virtual Labs, Ministry of Education, GOI, https://vlab.co.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620019	HS	--	--	3	1.5	50	50	3Hrs

English for Technical Communication and MOODLE Laboratory

(Common AIDS, AIML, CIC, CE, CSBS, CSD, CSE, CSIT, ECE, EEE, IT.)

Course Objectives:

1	Enable students apply correct phonological patterns for effective communication.
2	Develop students' ability to employ listening strategies to comprehend audio texts.
3	Enable students participate effectively in group discussions, presentations, debates, and interviews in professional contexts.
4	Enhance students' ability to write cohesively and coherently.
5	Familiarize students with Moodle LMS and digital learning tools for effective use of learning resources

Course Outcomes:

Sl. No	Outcome	Knowledge Level
1	Apply appropriate phonological features to communicate effectively in formal and informal contexts.	K3
2	Apply listening strategies to comprehend audio texts, take notes, and narrate events.	K3
3	Demonstrate effective participation in group discussions, presentations, debates, and interviews to communicate in professional contexts.	K3
4	Develop an ability to write clearly on topics related to technology and workplace communication.	K3
5	Demonstrate proficiency in using Moodle LMS and digital learning tools to access learning resources effectively.	K2

SYLLABUS

PART-A

1	Phonetics & Accent Neutralization
2	Introducing Self and Others
3	Brief Talk in Formal/Informal Contexts, JAM/Extempore
4	Listening Comprehension
5	Audio-based Note Taking & Narrating Events
6	Process Description and Paragraph Writing (Technical Lab Context)
7	Group Discussions and Deliberations
8	Micro Presentations (2-5 Mts)
9	Presenting Logical Arguments – Debates
10	Interviews – Formal & Semi-formal

PART-B Moodle Learning Management System: Digital Learning and Academic Skills	
1	Introduction to Moodle LMS • Introduction to Moodle • Student login and profile management • Moodle dashboard and course navigation • Accessing announcements and course materials • Notifications and calendars
2	Learning Through Moodle • Accessing PDFs, videos, web resources, and lecture materials • Downloading and organizing study materials • Moodle lessons and learning activities • Tracking course completion • Managing deadlines and study schedules
3	Assignments, Quizzes and Assessments • Understanding assignment instructions • Creating and uploading assignments • File formats and file naming • Online quizzes and question types • Viewing marks and teacher feedback • Resubmission and revision
4	Digital Communication and Collaboration • Moodle forums and discussions • Academic email etiquette • Online communication and netiquette • Group activities and peer learning • Collaborative learning tools • Digital presentation skills
Lab Manual:	
1	Propel: A Course Book for Technical English Communication. Cengage Publishers, 2026.
Reference Books:	
1	Richards, Jack C., Jonathan Hull, and Susan Proctor. Interchange Level 3 Student's Book with Self-study DVD-ROM. Vol. 3. Cambridge University Press, 2012.
2	Taylor Grant: English Conversation Practice,Tata McGraw Hill Education India,2016.
3	Hewing's, Martin. Cambridge Academic English (B2).CUP, 2012.
4	J.Sethi & P.V.Dhamija. A Course in Phonetics and Spoken English, (2ndEd), Kindle, 2013
5	Raman Meenakshi,Sangeeta-Sharma.TechnicalCommunication.OxfordPress.2018.

Web Resources:	
	1. www.esl-lab.com 2. www.englishmedialab.com 3. www.englishinteractive.net 4. https://www.britishcouncil.in/english/online 5. http://www.letstalkpodcast.com 6. https://www.youtube.com/c/mmmEnglish_Emma/featured 7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured 8. https://www.youtube.com/c/engvidAdam/featured 9. https://www.youtube.com/c/EnglishClass101/featured 10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists 11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw 12. https://www.youtube.com/user/letstalkaccent/videos 13. https://www.youtube.com/c/EngLanguageClub/featured 14. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc 15. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620020	MC	2	--	--	--	50	--	--

Universal Human Values – II Understanding Harmony and Ethical Human Conduct

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

Course Objectives: The objective of this course is to make the student aware of

1.	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2.	Harmony in the human being, family, society and nature/existence
3.	Holistic perspective towards life, profession and happiness

Course Outcomes: At the end of this course student will be able to

S.No	Outcome	Knowledge Level
1.	**Explain** the role of value education in achieving basic human aspirations.	K2
2.	**Summarize** needs to obtain harmony in self (I).	K2
3.	**Describe** criteria for human-human relationship and harmony in society.	K2
4.	**Explain** four orders of nature and our existence.	K2
5.	**Interpret** significance of harmony in holistic development	K2

SYLLABUS

UNIT-I (06Hrs)	**Introduction to Value Education**: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education); Understanding Value Education-Practice Session PS1 Sharing about Oneself; self-exploration as the Process for Value Education; Continuous Happiness and Prosperity – the Basic Human Aspirations-Practice Session PS2 Exploring Human Consciousness; Happiness and Prosperity – Current Scenario; Method to Fulfill the Basic Human Aspirations-Practice Session PS3 Exploring Natural Acceptance.
UNIT-II (08Hrs)	**Harmony in the Human Being**: Understanding Human being as the Co-existence of the self and the body; Distinguishing between the Needs of the self and the body -Practice Session PS4 Exploring the difference of Needs of self and body; The body as an Instrument of the self; Understanding Harmony in the self-Practice Session PS5 Exploring Sources of Imagination in the self; Harmony of the self with the body; Programme to ensure self-regulation and Health - Practice Session PS6 Exploring Harmony

	of self with the body.
UNIT-III (06 Hrs)	**Harmony in the Family and Society**: Harmony in the Family – the Basic Unit of Human Interaction; 'Trust' – the Foundational Value in Relationship-Practice Session PS7 Exploring the Feeling of Trust; 'Respect' – as the Right Evaluation-Practice Session PS8 Exploring the Feeling of Respect; Other Feelings, Justice in Human-to-Human Relationship; Understanding Harmony in the Society; Vision for the Universal Human Order- Practice Session PS9 Exploring Systems to fulfil Human Goal.
UNIT-IV (06 Hrs)	**Harmony in the Nature/Existence:** Understanding Harmony in the Nature; Interconnectedness, self- regulation and Mutual Fulfilment among the Four Orders of Nature-Practice Session PS10 Exploring the Four Orders of Nature; Realizing Existence as Co-existence at All Levels; The Holistic Perception of Harmony in Existence-Practice Session PS11 Exploring Co-existence in Existence.
UNIT-V (06 Hrs)	**Implications of the Holistic Understanding – a Look at Professional Ethics**: Natural Acceptance of Human Values; Definitiveness of (Ethical) Human Conduct- Practice Session PS12 Exploring Ethical Human Conduct; A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order; Competence in Professional Ethics-Practice Session PS13 Exploring Humanistic Models in Education; Holistic Technologies, Production Systems and Management Models-Typical Case Studies; Strategies for Transition towards Value-based Life and Profession-Practice Session PS14 Exploring Steps of Transition towards Universal Human Order.
Textbooks:	
1.	The Textbook R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1
2.	The Teacher's Manual R R Gaur, R Asthana, G P Bagaria,*Teachers' Manual for A Foundation Course in Human Values and Professional Ethics,* 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2
Reference Books:	

1.	*JeevanVidya: EkParichaya*, A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth – by Mohandas Karamchand Gandhi
5.	Small is Beautiful – E. F Schumacher.
6.	Slow is Beautiful – Cecile Andrews
7.	Economy of Permanence – J C Kumarappa
8.	Bharat Mein Angreji Raj – PanditSunderlal
9.	Rediscovering India – by Dharampal
10.	*Hind Swaraj or Indian Home Rule* – by Mohandas K. Gandhi
11.	*India Wins Freedom* – Maulana Abdul Kalam Azad
12.	*Vivekananda* – Romain Rolland (English)
13.	*Gandhi* – Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf
3.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf
4.	https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%2023.pdf
5.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf
6.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf
7.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
8.	https://www.studocu.com/in/document/kiet-group-of-institutions/universal-human-values/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385
9.	https://onlinecourses.swayam2.ac.in/aic22_ge23/preview